

Ideal and Perfect Secret Sharing for Fault-Tolerant Distributed Access Control of Secure Data via Nondominated Bicoterie

Jehn-Ruey Jiang
Department of Information Management
Hsuan Chuang University
Hsin-Chu, 300, Taiwan
jrjiang@hcu.edu.tw

Abstract

A perfect secret sharing scheme permits a secret to be shared among participants so that only qualified subsets of participants can reconstruct the secret, but any unqualified subset has absolutely no information about the secret. In this paper, we first show that the general secret sharing scheme with a coterie access structure is perfect if the coterie is nondominated (ND). We then propose a perfect secret sharing scheme with an ND bicoterie access structure for the access control of secure data. The proposed scheme has the merits of having two types of quorums: one type to grant and the other to revoke the access of the secure data. This provides us more flexibility for the distributed access control of the secure data. Since there are a large number of different quorums consisting of few participants, the proposed scheme is fault-tolerant in the sense that it still works even when some participants are not available. We also propose an ideal and perfect secret sharing scheme with an ND grid bicoterie access structure. The proposed scheme is optimal (ideal) in the sense that its shares have the same size as the secret.

Keywords: bicoterie, coterie nondomination, distributed access control, data security, fault-tolerance, secret sharing

1 Introduction

Secret sharing has received considerable attention since it was first introduced by Blakley [4] and Shamir [25]. It deals with the problem of how to securely share a secret among a group of participants and has many applications, such as secure data access control [21], secure multiparty computation [2], visual cryptography [20] and network auction [19], and so on. In a secret sharing scheme, a secret is divided into pieces, each of which is called a *share* and is held by a participant. A secret sharing scheme must ensure that only a qualified subset of participants can reconstruct the secret. If an unqualified subset of

participants cannot obtain any information of the secret from their shares, then the secret sharing scheme is called *perfect*. The *information rate* of a secret sharing scheme is defined to be the ratio of the secret size to the largest share size. It has been shown in [10] that any perfect secret sharing scheme must have information rate ≤ 1 . A perfect secret sharing scheme is said to be *ideal* if it has information rate of 1.

The collection of qualified subsets of participants is called an *access structure* [3, 13]. Many secret sharing schemes with different access structures have been proposed in the literature [3, 5-9, 13, 23, 26-27]. In [2, 17, 21], secret sharing schemes with *coterie* access structures are proposed. A coterie [12] is a collection of subsets of participants, where each subset is called a *quorum* and any pair of quorums must intersect. The coterie is extensively studied in the literature of distributed systems because it is very useful for solving the distributed mutual exclusion [1, 12, 14, 16, 18, 24] and replica control [15-16] problems, etc. In [21], secret sharing schemes with coterie access structures are further applied to granting and revoking the access of secure data, such as decryption keys and signatures, etc. By distributing shares of secure data over wide-spread servers, shares from a complete quorum of servers can restore the secure data. And after access authorization is revoked, an unauthorized user will never be able to restore the data even if some servers still consider him/her authorized.

A general secret sharing scheme with ordinary coterie access structure was proposed in [21]. However, as we will show later, this scheme is not always perfect. In this paper, to make the work in [21] more complete, we show that the general secret sharing scheme with the coterie access structure is perfect if the coterie is *nondominated* (ND). The ND coterie [12, 14, 22] is an important type of coterie, which we will introduce later.

Specific secret sharing schemes with many types of coterie access structures was also proposed in [2, 17, 21]. Those access structures are constructed on the basis of nondominated *wr-coterie* [17], the crumbling wall coterie [24],

the tree coterie [1], and the finite projective plane coterie [18], etc. All the abovementioned coterie are ND, so they all render the secret sharing scheme perfect. However, only the last two coterie render the secret sharing scheme *ideal*. A secret sharing scheme is called ideal if the size of shares is the same as that of the secret. It has been shown in [10] that all perfect secret sharing scheme must have share size as large as the secret size (i.e., information rate ≤ 1). Certainly, small shares are preferred because the security of the system degrades as the amount of share increases and small shares can be distributed and stored more efficiently.

In this paper, we first show that the general secret sharing scheme [21] with the coterie access structure is perfect if the coterie is ND. We then present a perfect secret sharing scheme with ND bicoterie access structures for the access control of secure data. A bicoterie $\mathcal{B}=(\mathcal{P}, \mathcal{Q})$ is an ordered pair of collections of sets, where each set is called a quorum and any quorum in $\mathcal{P}$ intersects any quorum in $\mathcal{Q}$. The secret sharing scheme with bicoterie access structures has the merits of having two types of quorums: one type to grant and the other to revoke the access of the secure data. This provides us more flexibility for the distributed access control of the secure data. Since there are a large number of different quorums consisting of few participants, the proposed scheme is fault-tolerant in the sense that it still works even when some participants are not available. At last, we propose an ideal and perfect secret sharing scheme with a specific access structure, which is constructed on the basis of an ND bicoterie — the grid bicoterie.

The rest of this paper is organized as follows. In Section 2, we elaborate some preliminaries about secret sharing schemes with the coterie and the bicoterie access structures. In Section 3, we introduce some theorems about perfect secret sharing schemes with the coterie and the bicoterie access structures. In Section 4, we propose an ideal and perfect secret sharing scheme with ND grid bicoterie access structure. Finally, we give some concluding remarks in Section 5.

2 Preliminaries

Let $U=\{u_1, \dots, u_n\}$ be the universal set of all participants $u_1, \dots, u_n$. Below, we introduce the definitions of coterie, bicoterie and the nondominated (ND) concept of them. We also introduce the preliminaries about secret sharing schemes with access structures constructed on the basis of coterie and bicoterie.

Definition 1. Coterie

A coterie [12] $\mathcal{C}$ is a collection of subsets of U , where each subset is called a quorum satisfying the following properties:

Intersection Property: $\forall G, H \in \mathcal{C}: G \cap H \neq \emptyset$;

Minimality Property: $\forall G, H \in \mathcal{C}: G \not\subseteq H$.

Definition 2. Bicoterie

A bicoterie [11] $\mathcal{B}$ is an ordered pair $(\mathcal{P}, \mathcal{Q})$, where $\mathcal{P}$ and $\mathcal{Q}$ are collections of subsets (called quorums) of U . A bicoterie $\mathcal{B}$ should satisfy the following properties:

Intersection Property: $\forall G \in \mathcal{P}, H \in \mathcal{Q}: G \cap H \neq \emptyset$;

Minimality Property: $[\forall G, H \in \mathcal{P}: G \not\subseteq H]$ and $[\forall G, H \in \mathcal{Q}: G \not\subseteq H]$.

For example, let $\mathcal{P}=\{\{u_1, u_2, u_3\}, \{u_1, u_2, u_4\}, \{u_3, u_4\}\}$ and $\mathcal{Q}=\{\{u_1, u_3\}, \{u_1, u_4\}, \{u_2, u_3\}, \{u_2, u_4\}, \{u_3, u_4\}\}$. Then, $(\mathcal{P}, \mathcal{Q})$ is a bicoterie since it satisfies the intersection and the minimality properties.

Definition 3. Domination of Coterie

Let $\mathcal{C}_1$ and $\mathcal{C}_2$ be two coterie. We say that $\mathcal{C}_2$ dominates $\mathcal{C}_1$ if and only if

- (1) $\mathcal{C}_1 \neq \mathcal{C}_2$;
- (2) $\forall G \in \mathcal{C}_1, \exists H \in \mathcal{C}_2: H \subseteq G$.

Definition 4. Domination of Bicoterie

Let $\mathcal{B}_1=(\mathcal{P}_1, \mathcal{Q}_1)$ and $\mathcal{B}_2=(\mathcal{P}_2, \mathcal{Q}_2)$ be two bicoterie. We say $\mathcal{B}_2$ dominates $\mathcal{B}_1$ if and only if

- (1) $\mathcal{B}_1 \neq \mathcal{B}_2$, i.e., $\mathcal{P}_1 \neq \mathcal{P}_2$ and/or $\mathcal{Q}_1 \neq \mathcal{Q}_2$;
- (2) $\forall G \in \mathcal{P}_1, \exists H \in \mathcal{P}_2: H \subseteq G$;
- (3) $\forall G \in \mathcal{Q}_1, \exists H \in \mathcal{Q}_2: H \subseteq G$.

For example, let $\mathcal{P}_1=\{\{u_1, u_2, u_3\}, \{u_1, u_2, u_4\}, \{u_1, u_3, u_4\}, \{u_2, u_3, u_4\}\}$, $\mathcal{Q}_1=\{\{u_1, u_3\}, \{u_1, u_4\}, \{u_2, u_3\}, \{u_2, u_4\}\}$, $\mathcal{P}_2=\{\{u_1, u_2, u_3\}, \{u_1, u_2, u_4\}, \{u_3, u_4\}\}$, and $\mathcal{Q}_2=\{\{u_1, u_3\}, \{u_1, u_4\}, \{u_2, u_3\}, \{u_2, u_4\}, \{u_3, u_4\}\}$. Then, $\mathcal{B}_1=(\mathcal{P}_1, \mathcal{Q}_1)$ and $\mathcal{B}_2=(\mathcal{P}_2, \mathcal{Q}_2)$ are bicoterie and $\mathcal{B}_2$ dominates $\mathcal{B}_1$.

Definition 5. Nondomination of Coterie and Bicoterie

A coterie (bicoterie) is *nondominated* (ND) if and only if no other coterie (bicoterie) can dominate it.

Definition 6. Access Structures

An access structure Γ is the collection of subsets of U , which satisfying the monotone increasing property, that is, if $G \in \Gamma$ and $G \subseteq H$, then we have $H \in \Gamma$.

Definition 7. Coterie Access Structures

Let $\mathcal{C}$ be a coterie. The coterie access structure based on $\mathcal{C}$, which is denoted with $\Gamma(\mathcal{C})$, is defined as the collection of all supersets of quorums in $\mathcal{C}$. That is, $\Gamma(\mathcal{C}) = \{H \mid H \supseteq G, G \in \mathcal{C}\}$.

Definition 8. Bicoterie Access Structures

Let $\mathcal{B} = (\mathcal{P}, \mathcal{Q})$ be a bicoterie. The bicoterie access structure based on $\mathcal{B}$, which is denoted with $\Gamma(\mathcal{B})$, is defined as the collection of all supersets of quorums in $\mathcal{P}$. That is, $\Gamma(\mathcal{B}) = \{H \mid H \supseteq G, G \in \mathcal{P}\}$. Note that $\Gamma(\mathcal{B}) = \Gamma(\mathcal{P})$ and we will use this fact in the theorem proofs in the next section.

Definition 9. Secret Sharing Schemes

Let S be the domain of all possible secrets. A secret sharing scheme is a mapping $\Pi: S \times E \rightarrow S_1 \times \dots \times S_n$, where E is the domain of random numbers for splitting the secret, and S_i is the domain of shares distributed to participant u_i , where u_i belongs to $U = \{u_1, \dots, u_n\}$, the universal set of all participants. We say that Π is a secret sharing scheme with access structure Γ if the following reconstruction property is satisfied:

(S1) Reconstruction Property:

Let $H = \{u_{\sigma(1)}, \dots, u_{\sigma(k)}\}$ be any qualified subset of participants in Γ , where $\sigma(1), \dots, \sigma(k)$ are distinct and $1 \leq \sigma(1), \dots, \sigma(k) \leq n$. Associated with H , there is a function $F_H: S_{\sigma(1)} \times \dots \times S_{\sigma(k)} \rightarrow S$, such that for every pair $(s, \varepsilon) \in S \times E$, if $\Pi(s, \varepsilon) = (s_1, \dots, s_n)$ then $F_H(s_{\sigma(1)}, \dots, s_{\sigma(k)}) = s$. That is, the secret can be reconstructed by shares from any subset of Γ .

Further, Π is said to be *perfect* if the following property is satisfied:

(S2) Perfect Property:

Let $H \notin \Gamma$ be an unqualified subset of participants. And let a and $b \in S$ be any two secrets. For every possible collection of shares $\{s_i \mid u_i \in H\}$, we have $\text{prob}(a \mid \{s_i \mid u_i \in H\}) = \text{prob}(b \mid \{s_i \mid u_i \in H\})$. That is, from the set of shares of unqualified subset of participants, the probability of deriving the secret to be a , and the probability of deriving the secret to be b is equal, which implies that no information of the secret is revealed.

3 Secret Sharing Schemes with Coterie and Bicoterie Access Structures

In this section, we first show that the general secret sharing scheme with coterie access structure $\Gamma(\mathcal{C})$ is perfect if $\mathcal{C}$ is ND to make the work in [21] more complete. We then propose a perfect secret sharing scheme with ND bicoterie access structures.

3.1 The Secret Sharing Scheme with Coterie Access Structures

In [21], a general secret sharing scheme with coterie access structure $\Gamma(\mathcal{C})$ was proposed. This scheme can be used to distribute access control of the secure data so that after access authorization is revoked, an unauthorized user will never be able to restore the data even if some servers still consider him/her authorized. Since there are a large number of different quorums consisting of few participants, the proposed scheme is fault-tolerant in the sense that it still works even when some participants are not available. However, the secret sharing scheme mentioned is not perfect in the sense that even an unauthorized subset of participants may reconstruct the secret if $\mathcal{C}$ is dominated. Below, we show that the secret sharing scheme with coterie access structure $\Gamma(\mathcal{C})$ is perfect if $\mathcal{C}$ is ND.

We first describe the secret sharing scheme Π proposed in [21] as follows. Given a coterie $\mathcal{C} = \{G_1, \dots, G_m\}$ and a secret $v \in S$. We can split v into $v_1, \dots, v_m$ randomly according to a random number $\varepsilon \in E$ such that $v = v_1 + \dots + v_m$. We call each of $v_1, \dots, v_m$ the *partition* of v . Each v_j , $1 \leq j \leq m$, is associated with a quorum $G_j \in \mathcal{C}$. The shares $s_i(v) \in S_i$ of secret v distributed to participant u_i is defined as the set of all partitions associated with quorums that contain u_i . More precisely, $s_i(v) = \{v_j \mid u_i \in G_j, 1 \leq j \leq m\}$ for $1 \leq i \leq n$. In [21], the mapping $\Pi(v, \varepsilon) = (s_1(v), \dots, s_n(v))$ of $S \times E \rightarrow S_1 \times \dots \times S_n$ is shown to be a secret sharing scheme with the coterie access structure $\Gamma(\mathcal{C})$. Below, we further show that Π is perfect if $\mathcal{C}$ is ND.

We first introduce a theorem in [12], which is concerned with the properties of nondominated coterie.

Theorem 1. Let $\mathcal{C}$ be a coterie under the universal set U . $\mathcal{C}$ is dominated if and only if there exists a subset H of U such that

- 1) $\forall G \in \mathcal{C}: G \not\subseteq H$.
- 2) $\forall G \in \mathcal{C}: G \cap H \neq \emptyset$.

Based on theorem 1, we prove theorem 2 below. Before proceeding to theorem 2, we remind the reader to note the following facts: $[G \cap H = \emptyset \text{ iff } G \subseteq \overline{H}]$ and $[G \cap H = \emptyset \text{ iff } H \subseteq \overline{G}]$, or equivalently, $[G \cap H \neq \emptyset \text{ iff } G \not\subseteq \overline{H}]$ and $[G \cap H \neq \emptyset \text{ iff } H \not\subseteq \overline{G}]$.

Theorem 2. Let $\mathcal{C}$ be a coterie under the universal set U , and let $\Gamma(\mathcal{C})$ be the access structure based on $\mathcal{C}$. It follows that $\mathcal{C}$ is dominated if and only if there exists a subset H of U such that $H \notin \Gamma(\mathcal{C})$ and $\overline{H} \notin \Gamma(\mathcal{C})$.

Proof:

(if part) $H \notin \Gamma(\mathcal{C})$ implies $\forall G \in \mathcal{C}: G \not\subseteq H$. And $\overline{H} \notin \Gamma(\mathcal{C})$ implies $\forall G \in \mathcal{C}: G \not\subseteq \overline{H}$, which in turn implies $\forall G \in \mathcal{C}: G \cap H \neq \emptyset$. By theorem 1, we have that $\mathcal{C}$ is dominated.

(only if part) By theorem 1, if $\mathcal{C}$ is dominated, then there exists a subset H of U such that $\forall G \in \mathcal{C}: G \not\subseteq H$, which implies $H \notin \Gamma(\mathcal{C})$. By theorem 1, if $\mathcal{C}$ is dominated, then $\forall G \in \mathcal{C}: G \cap H \neq \emptyset$, which implies $\forall G \in \mathcal{C}: G \not\subseteq \overline{H}$; i.e., $\overline{H} \notin \Gamma(\mathcal{C})$. $\square$

Below, we introduce the following corollaries, which are direct consequences of theorem 2.

Corollary 1. Let $\mathcal{C}$ be a coterie under the universal set U , and let $\Gamma(\mathcal{C})$ be the access structure based on $\mathcal{C}$. $\mathcal{C}$ is ND if and only $G \in \Gamma(\mathcal{C})$ or $\overline{G} \in \Gamma(\mathcal{C})$ for every subset G of U .

Corollary 2. Let $\mathcal{C}$ be a coterie under the universal set U , and let $\Gamma(\mathcal{C})$ be the access structure based on $\mathcal{C}$. If $\mathcal{C}$ is ND then $[G \in \Gamma(\mathcal{C}) \text{ implies } \overline{G} \notin \Gamma(\mathcal{C})]$, $[G \in \Gamma(\mathcal{C}) \text{ implies } \overline{G} \notin \Gamma(\mathcal{C})]$, $[G \notin \Gamma(\mathcal{C}) \text{ implies } \overline{G} \in \Gamma(\mathcal{C})]$, and $[G \notin \Gamma(\mathcal{C}) \text{ implies } \overline{G} \in \Gamma(\mathcal{C})]$.

We can now prove that Π is a perfect secret sharing scheme with the coterie access structure $\Gamma(\mathcal{C})$ if $\mathcal{C}$ is ND.

Theorem 3. Let Π be a mapping as described earlier. If $\mathcal{C}$ is ND, then Π is a perfect secret sharing scheme with the coterie access structure $\Gamma(\mathcal{C})$.

Proof :

The reconstruction property of Π has been shown in [21], below we show its perfect property:

Let $H = \{u_{\sigma(1)}, \dots, u_{\sigma(k)}\} \notin \Gamma(\mathcal{C})$, where $\sigma(1), \dots, \sigma(k)$ are distinct and $1 \leq \sigma(1), \dots, \sigma(k) \leq n$. Because $\mathcal{C}$ is ND, we have $\overline{H} \in \Gamma(\mathcal{C})$ by corollary 2. It follows that there exists a quorum $G_j \in \mathcal{C}$ such that $G_j \subseteq \overline{H}$, which in turn implies $G_j \cap H = \emptyset$. Only the members of G_j own the partition v_j . So, no participant in H obtains v_j .

Let a and b be two different secrets. And let $\Pi(a, \mathcal{E}^a) = (s_1^a, \dots, s_n^a)$ be a secret split of a according to $\mathcal{E}^a$ such that $F_H(s_{\sigma(1)}^a, \dots, s_{\sigma(k)}^a) = a$. We can form another secret split $\Pi(b, \mathcal{E}^b) = (s_1^b, \dots, s_n^b)$ such that $F_H(s_{\sigma(1)}^b, \dots, s_{\sigma(k)}^b) = b$. The secret split $(s_1^b, \dots, s_n^b)$ can be derived from $(s_1^a, \dots, s_n^a)$ by substituting the partition v_j with $v_j + (a - b)$. Since $G_j \cap H = \emptyset$ and the substitution of v_j only affect the participants in G_j , the projection of $(s_1^a, \dots, s_n^a)$ and $(s_1^b, \dots, s_n^b)$ on participants of H are identical. So, we have

$prob(a | \{s_i(v) | u_i \in H\}) = prob(b | \{s_i(v) | u_i \in H\})$. That means that the shares from participants in H can not reveal any information of the secret value v .

$\square$

3.2 The secret sharing scheme with ND bicoterie access structures

Given a bicoterie $\mathcal{B} = (\mathcal{P}, \mathcal{Q})$ and a secret value $v \in S$, where $\mathcal{P} = \{H_1, \dots, H_l\}$ and $\mathcal{Q} = \{G_1, \dots, G_m\}$. We can split v into $v_1, \dots, v_m$ randomly according to a random number $\varepsilon \in E$, such that $v = v_1 + \dots + v_m$. We call each of $v_1, \dots, v_m$ the *partition* of v . Each v_j , $1 \leq j \leq m$, is associated with a quorum $G_j \in \mathcal{Q}$. The shares $s_i(v) \in S_i$ of secret v distributed to participant u_i is defined as the set of all partitions associated with $\mathcal{Q}$'s quorums that contain u_i . More precisely, $s_i(v) \equiv \{v_j | u_i \in G_j, 1 \leq j \leq m\}$ for $1 \leq i \leq n$. Below, we show that $\Sigma(v, \varepsilon) = (s_1(v), \dots, s_n(v))$ of $S \times E \rightarrow S_1 \times \dots \times S_n$ is a perfect secret sharing scheme with the bicoterie access structure $\Gamma(\mathcal{B})$ if $\mathcal{B}$ is ND.

Below, we introduce a theorem [22] about ND bicoterie.

Theorem 4. Let $\mathcal{B} = (\mathcal{P}, \mathcal{Q})$ be a bicoterie under the universal set U . $\mathcal{B}$ is dominated if and only if there exists a subset H of U such that

- 1). $\forall G \in \mathcal{P}: G \not\subseteq H$
- 2). $\forall G \in \mathcal{Q}: G \cap H \neq \emptyset$.

On the basis of theorem 4, we prove the following theorem.

Theorem 5. Let $\mathcal{B} = (\mathcal{P}, \mathcal{Q})$ be a bicoterie under the universal set U . Also let $\Gamma(\mathcal{P})$ and $\Gamma(\mathcal{Q})$ be the access structures based on $\mathcal{P}$ and $\mathcal{Q}$, respectively. It follows that $\mathcal{B}$ is dominated if and only if there exists a subset H of U such that $H \notin \Gamma(\mathcal{P})$ and $\overline{H} \notin \Gamma(\mathcal{Q})$.

Proof:

(if part) $H \notin \Gamma(\mathcal{P})$ implies $\forall G \in \mathcal{P}: G \not\subseteq H$. And $\overline{H} \notin \Gamma(\mathcal{Q})$ implies $\forall G \in \mathcal{Q}: G \not\subseteq \overline{H}$, which in turn implies $\forall G \in \mathcal{Q}: G \cap H \neq \emptyset$. By theorem 4, we have that $\mathcal{B}$ is dominated.

(only if part) By theorem 4, if $\mathcal{B}$ is dominated, then there exists a subset H of U such that $\forall G \in \mathcal{P}: G \not\subseteq H$, which implies $H \notin \Gamma(\mathcal{P})$. By theorem 4, we also have $\forall G \in \mathcal{Q}: G \cap H \neq \emptyset$, which implies $\forall G \in \mathcal{Q}: G \not\subseteq \overline{H}$; i.e., $\overline{H} \notin \Gamma(\mathcal{Q})$. $\square$

Below, we introduce the following corollaries, which are direct consequences of theorem 5.

Corollary 3. Let $\mathcal{B} = (\mathcal{P}, \mathcal{Q})$ be a bicoterie under the universal set U . Also let $\Gamma(\mathcal{P})$ and $\Gamma(\mathcal{Q})$ be the access structures based on $\mathcal{P}$ and $\mathcal{Q}$, respectively.

$\mathcal{B}$ is ND if and only if $G \in \Gamma(\mathcal{P})$ or $\overline{G} \in \Gamma(\mathcal{Q})$ for every subset G of U .

Corollary 4. Let $\mathcal{B}=(\mathcal{P}, \mathcal{Q})$ be a bicoterie under the universal set U . Also let $\Gamma(\mathcal{P})$ and $\Gamma(\mathcal{Q})$ be the access structures based on $\mathcal{P}$ and $\mathcal{Q}$, respectively. If $\mathcal{B}$ is ND then $[G \in \Gamma(\mathcal{P}) \text{ implies } \overline{G} \notin \Gamma(\mathcal{Q})]$, $[G \in \Gamma(\mathcal{P}) \text{ implies } \overline{G} \notin \Gamma(\mathcal{Q})]$, $[G \notin \Gamma(\mathcal{P}) \text{ implies } \overline{G} \in \Gamma(\mathcal{Q})]$, and $[G \notin \Gamma(\mathcal{P}) \text{ implies } \overline{G} \in \Gamma(\mathcal{Q})]$.

We can now prove that Σ is a perfect secret sharing scheme with the bicoterie access structure $\Gamma(\mathcal{B})$ if $\mathcal{B}$ is ND.

Theorem 6. Let Σ be a mapping as described earlier. If $\mathcal{B}=(\mathcal{P}, \mathcal{Q})$ is an ND bicoterie, then Σ is a perfect secret sharing scheme with the bicoterie access structure $\Gamma(\mathcal{B})$.

Proof :

(1) Reconstruction Property:

Let $H \in \Gamma(\mathcal{B}) = \Gamma(\mathcal{P})$. According to the intersection property of $\mathcal{B}$, we have that H intersects with every quorum in $\mathcal{Q}$. So, the participants in H can collect all partitions $v_1, \dots, v_m$ of v and reconstruct the secret value v by $v = v_1 + \dots + v_m$, where m is the cardinality of $\mathcal{Q}$.

(2) Perfect Property:

Let $H = \{u_{\sigma(1)}, \dots, u_{\sigma(k)}\} \in \Gamma(\mathcal{B}) = \Gamma(\mathcal{P})$, where $\sigma(1), \dots, \sigma(k)$ are distinct and $1 \leq \sigma(1), \dots, \sigma(k) \leq n$. Because $\mathcal{B}$ is ND, we have $\overline{H} \in \Gamma(\mathcal{Q})$ by corollary 4. It follows that there exists a quorum $G_j \in \mathcal{Q}$ such that $G_j \subseteq \overline{H}$, which in turn implies $G_j \cap H = \emptyset$. Only the members of G_j own the partition v_j . So, no participant in H obtains v_j .

Let a and b be two different secrets. And let $\Sigma(a, \mathcal{E}^a) = (s_1^a, \dots, s_n^a)$ be a secret split of a according to $\mathcal{E}^a$ such that $F_H(s_{\sigma(1)}^a, \dots, s_{\sigma(k)}^a) = a$. We can form another secret split $\Sigma(b, \mathcal{E}^b) = (s_1^b, \dots, s_n^b)$ such that $F_H(s_{\sigma(1)}^b, \dots, s_{\sigma(k)}^b) = b$. The secret split $(s_1^b, \dots, s_n^b)$ can be derived from $(s_1^a, \dots, s_n^a)$ by substituting the partition v_j with $v_j + (a - b)$. Since $G_j \cap H = \emptyset$ and the substitution of v_j only affect the participants in G_j , the projection of $(s_1^a, \dots, s_n^a)$ and $(s_1^b, \dots, s_n^b)$ on participants of H are identical. So, we have $\text{prob}(a | \{s_i(v) | u_i \in H\}) = \text{prob}(b | \{s_i(v) | u_i \in H\})$. That means that the shares from participants in H can not reveal any information of the secret value v . $\square$

4 An ideal and perfect secret sharing scheme with grid bicoterie access structures

In this section, we show an ideal and perfect secret sharing scheme with the grid bicoterie access structure. Below, we first introduce the construction of grid bicoterie.

Given an universal set U of participants $u_1, \dots, u_n$, we organize participants into an h -row, w -column grid, where $n = hw$ (h stands for the height of grid, and w , the width). Note that for the case of $n \neq hw$, we can just rule out the grid points that are not associated with any participant. Figure 1 is the illustration of organizing 12 participants into a 3-row, 4-column grid.

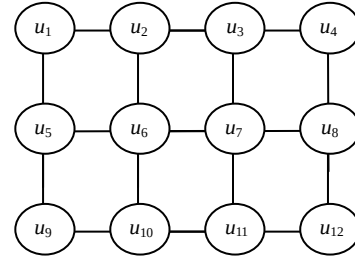


Figure 1. The illustration of organizing 12 participants into a 3-row, 4-column grid.

With the aid of the grid structure, the *grid bicoterie* $\mathcal{B}$ is defined to be an ordered pair $(\mathcal{P}, \mathcal{Q})$, where $\mathcal{Q}$ is a collection of all columns, and $\mathcal{P}$ is a collection of sets, each containing one participant from every column. Taking the grid structure of figure 1 for example, $\{u_1, u_5, u_9\}$ is a member of $\mathcal{Q}$, and $\{u_1, u_6, u_7, u_{12}\}$ is a member of $\mathcal{P}$.

In addition to the grid bicoterie, there are also ways to construct other bicoterie, which are called *extended grid bicoterie* in the following context. Below, we present two types of extended grid bicoterie:

(1) Type-1 Extended Grid Bicoterie $\mathcal{B}_1 = (\mathcal{P}_1, \mathcal{Q}_1)$: $\mathcal{Q}_1$ is the collection of sets, each containing r , $r \leq h$, participants from any column; $\mathcal{P}_1$ is the collection of sets, each containing $h - r + 1$ participants from every column.

Taking the grid structure in figure 1 for examples, for $r=2$, $\{u_2, u_6\}$ is a quorum of $\mathcal{Q}_1$ and $\{u_1, u_2, u_4, u_6, u_7, u_8, u_9, u_{11}\}$ is a quorum of $\mathcal{P}_1$. For $r=3$, $\{u_1, u_5, u_9\}$ is a quorum of $\mathcal{Q}_1$ and $\{u_1, u_6, u_7, u_{12}\}$ is a quorum of $\mathcal{P}_1$. Note that when $r=h$, a type-1 extended grid bicoterie is actually a grid bicoterie.

(2) Type-2 Extended Grid Bicoterie $\mathcal{B}_2 = (\mathcal{P}_2, \mathcal{Q}_2)$: $\mathcal{Q}_2$ is the collection of sets, each containing all participants from c , $c < w$, full columns; $\mathcal{P}_2$ is the collection of sets, each containing one participant

from $w-c+1$ columns.

Taking the grid structure in figure 1 for examples, for $c=2$, $\{u_2, u_4, u_6, u_8, u_{10}, u_{12}\}$ is a quorum of $\mathcal{Q}_2$ and $\{u_1, u_2, u_3\}$ is a quorum of $\mathcal{P}_2$. For $c=1$, $\{u_1, u_5, u_9\}$ is a quorum of $\mathcal{Q}_2$ and $\{u_1, u_6, u_7, u_{12}\}$ is a quorum of $\mathcal{P}_2$. Note that when $c=1$, a type-2 extended grid bicoterie is actually a grid bicoterie.

It is very easy to see $\mathcal{B}=(\mathcal{P}, \mathcal{Q})$, $\mathcal{B}_1=(\mathcal{P}_1, \mathcal{Q}_1)$ and $\mathcal{B}_2=(\mathcal{P}_2, \mathcal{Q}_2)$ are all bicoterie since they all satisfy the minimality property and every member of $\mathcal{P}$, $\mathcal{P}_1$ and $\mathcal{P}_2$ intersects a member of $\mathcal{Q}$, $\mathcal{Q}_1$ and $\mathcal{Q}_2$, respectively. Below, we introduce a theorem proposed in [22], which is useful to prove that the grid and the extended grid bicoterie are ND. Before proceeding to this theorem, we first define some terms about bicoterie:

Let $\mathcal{Q}$ be a collection of subsets of U . A *transversal* of $\mathcal{Q}$ is a subset of U that intersects with all members of $\mathcal{Q}$. That is, if $H \subseteq U$ is a transversal of $\mathcal{Q}$, then $\forall G \in \mathcal{Q}: H \cap G \neq \emptyset$. The set of all minimal transversals of $\mathcal{Q}$ is denoted by $\text{Tr}(\mathcal{Q})$.

Theorem 7. Let $\mathcal{B}=(\mathcal{P}, \mathcal{Q})$ be a bicoterie. Then, $\mathcal{B}$ is ND if and only if $\mathcal{P}$ is $\text{Tr}(\mathcal{Q})$.

On the basis of theorem 7, the reader can easily check that the grid and the extended grid bicoterie are ND (the proofs are trivial and omitted). Thus, they can all be applied to implement perfect secret sharing schemes. Below, we further show that the grid bicoterie can be applied to implement ideal and perfect secret sharing schemes.

Theorem 8. Let $\mathcal{B}=(\mathcal{P}, \mathcal{Q})$ be a grid bicoterie and let Σ be a function as defined earlier. Then, Σ is an ideal and perfect secret sharing scheme with the bicoterie access structure $\Gamma(\mathcal{B})$.

Proof:

Since the grid bicoterie is ND, Σ is a perfect secret sharing scheme with the bicoterie access structure $\Gamma(\mathcal{B})$ by theorem 6. Let u be a participant. Because u belongs to exactly one quorum G of $\mathcal{Q}$, the share of u contains only the partition associated with G , which is of the same size (the same domain) as the secret value. $\square$

5 Conclusion

In this paper, we have shown that the general secret sharing scheme proposed in [21] with the coterie access structure $\Gamma(\mathcal{C})$ is perfect if $\mathcal{C}$ is ND. We have also presented a secret sharing scheme with the bicoterie access structure for the access control of secure data. We have shown that the

grid bicoterie and the extended grid bicoterie can make the proposed secret sharing scheme perfect. We have also shown that the grid bicoterie further makes the proposed secret sharing scheme ideal. As we have mentioned, the secret sharing scheme with bicoterie access structures has the merits of having two types of quorums: one type to grant and the other to revoke the access of the secure data. This provides us more flexibility for the access control of the secure data. Since there are a large number of different quorums consisting of few participants, the proposed scheme is fault-tolerant in the sense that it still works even when some participants are not available. By grid bicoterie and extended grid bicoterie, we have even more flexibility on secure data access control in the sense that we can adjust h and w (the number of rows and columns of the grid) according to security requirements. For example, we can adjust w to be larger so that more pieces of shares are needed to reconstruct the secret value, and meanwhile, fewer servers can revoke the access authorization. In the future, we plan to investigate the effects of choosing different h and w and to investigate possible applications of secret sharing schemes with bicoterie access structures.

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